

NETWORKS AND COMPLEXITY

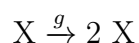
Solution 12-5

*This is an example solution from the forthcoming book *Networks and Complexity*.*

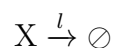
Find more exercises at <https://github.com/NC-Book/NCB>

Ex 12.5: Birth and death [Lvl. 3]

A population of bacteria X grows by cell division at rate g



In addition the bacteria die spontaneously at rate l



Derive an ODE for the population size x and solve it.

Solution

Both processes are proportional to X and increase / reduce the number of bacteria by 1 respectively, hence

$$\dot{x} = gx - lx = (g - l)x. \tag{1}$$

We already solved such a linear equation in the chapter. The solution is

$$x(t) = x_0 e^{(g-l)t}. \tag{2}$$